Stage 1 Hard Drill — With Solutions

Chapter 5.1 — Joint, Marginal & Conditional Probability Distributions

10 new competition-grade questions, full-chapter coverage, full worked solutions. Target: under 90 seconds each once the shortcut is spotted.

Q1 — Recovering a Missing Cell, Then Both Marginals

A joint PMF table for (X, Y), $X \in \{0,1\}$, $Y \in \{0,1,2\}$, has one unknown entry p:

	$Y{=}0$	$Y{=}1$	$Y{=}2$
$X{=}0$	0.08	0.12	0.05
$X{=}1$	0.15	p	0.25

Find p, then f_X and f_Y.

Solution. All six cells sum to 1: $0.08 + 0.12 + 0.05 + 0.15 + 0.25 = 0.65 \Rightarrow p = \boxed{0.35}$.

$$f_X(0) = 0.25,\ f_X(1) = 0.75$$

$$f_Y(0) = 0.23,\ f_Y(1) = 0.47,\ f_Y(2) = 0.30$$

⋆ *Intuition:* exactly Example 5's logic, one dimension bigger — a 2×3 table has 6 cells but only 5 independent numbers; the sixth is forced by the global sum-to-1 constraint, never extra information.

Q2 — Normalizing Constant on a Non-Square Rectangle

$f_{X,Y}(x,y) = c(2x+y)$ on the rectangle $0 < x < 1$, $0 < y < 2$. Find c and $f_X(x)$.

Solution.

$$\int_0^1 \int_0^2 c(2x+y)\,dy\,dx = c\int_0^1 (4x+2)\,dx = c(2+2) = 4c$$

Set $4c = 1 \Rightarrow c = \boxed{\frac{1}{4}}$.

$$f_X(x) = \int_0^2 \tfrac{1}{4}(2x+y)\,dy = \tfrac{1}{4}(4x+2) = \boxed{x + \tfrac{1}{2}}$$

for $0 < x < 1$.

▶ *Speed trick:* the y-range being $(0,2)$ instead of $(0,1)$ doesn't change the method, only the arithmetic — always integrate the un-normalized shape over its *actual* support first, never assume a unit interval.

Q3 — The Near-Miss Independence Trap

$X, Y \in \{1,2\}$ with joint PMF $f(1,1) = 0.06$, $f(1,2) = 0.14$, $f(2,1) = 0.25$, $f(2,2) = 0.55$. Are X, Y independent?

Solution. Marginals: $f_X(1) = 0.20$, $f_X(2) = 0.80$; $f_Y(1) = 0.31$, $f_Y(2) = 0.69$. Test the riskiest cell:

$$f_X(2) f_Y(2) = 0.80 \times 0.69 = 0.552 \neq 0.55 = f(2,2)$$

Off by just 0.002, but that's still a mismatch: $\boxed{\text{dependent}}$.

⋆ *Intuition:* a near-miss is still a miss — competition answers must be checked to exact decimal/fraction precision, not "looks close enough." This is exactly why Example 7's one-cell trap matters: always verify *every* cell symbolically (as a fraction) rather than eyeballing rounded decimals.

Q4 — Confirming Independence, Continuous Case

$f_{X,Y}(x,y) = 6x^2 y$ on $[0,1] \times [0,1]$. Are X, Y independent?

Solution. Normalization: $\int_0^1 \int_0^1 6x^2 y\,dy\,dx = \int_0^1 3x^2\,dx = 1$ ✓. Marginals: $f_X(x) = \int_0^1 6x^2 y\,dy = 3x^2$; $f_Y(y) = \int_0^1 6x^2 y\,dx = 2y$ (both valid densities on $[0,1]$). Check factorization:

$$f_X(x) f_Y(y) = (3x^2)(2y) = 6x^2 y = f_{X,Y}(x,y) \quad \text{for every } (x,y)$$

$\boxed{\text{Independent.}}$

▶ *Speed trick:* $f(x,y) = 6x^2 y$ is already visibly of the form $g(x)h(y)$ on a rectangle — you can declare independence on sight, then compute the marginals only to report their exact form, not to test anything further.

Q5 — Conditional PMF From Q1's Table

Using Q1's completed table, find $f_{Y|X}(y \mid X{=}1)$.

Solution. Row $X = 1$: $(0.15, 0.35, 0.25)$, divide by $f_X(1) = 0.75$:

$$f_{Y|X}(0 \mid 1) = \tfrac{0.15}{0.75} = \tfrac{1}{5}, \quad f_{Y|X}(1 \mid 1) = \tfrac{0.35}{0.75} = \tfrac{7}{15}$$

$$f_{Y|X}(2 \mid 1) = \tfrac{0.25}{0.75} = \tfrac{1}{3}$$

Check: $\frac{1}{5} + \frac{7}{15} + \frac{1}{3} = \frac{3}{15} + \frac{7}{15} + \frac{5}{15} = 1$ ✓.

⋆ *Intuition:* zoom into row $X{=}1$ and re-normalize — the row's raw weights $(0.15, 0.35, 0.25)$ don't sum to 1 on their own, so dividing by $f_X(1) = 0.75$ rescales them into a genuine standalone distribution over Y.

Q6 — A New Triangle, Recognized as Uniform

$f_{X,Y}(x,y) = 1$ on the triangle $\{x > 0,\ y > 0,\ 2y + x < 2\}$ (vertices $(0,0), (2,0), (0,1)$). Find $f_X(x)$, identify $Y \mid X{=}x$ by name, and find $\mathbb{E}[Y \mid X = x]$.

Solution. For fixed $x \in (0,2)$, y ranges over $\left(0, 1 - \frac{x}{2}\right)$:

$$f_X(x) = \int_0^{1-x/2} 1\,dy = \boxed{1 - \tfrac{x}{2}}, \quad 0 < x < 2$$

$$f_{Y|X}(y \mid x) = \frac{1}{1 - x/2} \Rightarrow \boxed{Y \mid X{=}x \sim \text{Uniform}\left(0,\ 1 - \tfrac{x}{2}\right)}$$

$$\mathbb{E}[Y \mid X = x] = \frac{1 - x/2}{2} = \boxed{\frac{2-x}{4}}, \quad \text{so } \mathbb{E}[Y \mid X{=}1] = \tfrac{1}{4}$$

▶ *Speed trick:* any uniform joint density on a triangle conditions down to a uniform conditional density on the shrunken segment — the instant you see a flat density on a triangular support, skip straight to "conditional is Uniform(0, ceiling(x))" and quote its mean as ceiling/2.

Q7 — Tail Probability From Q6's Conditional

Using Q6's conditional law, find $\mathbb{P}(Y > 0.4 \mid X = 0.5)$.

Solution. At $x = 0.5$: $Y \mid X{=}0.5 \sim \text{Uniform}(0,\ 1 - 0.25) = \text{Uniform}(0, 0.75)$.

$$\mathbb{P}(Y > 0.4 \mid X = 0.5) = \frac{0.75 - 0.4}{0.75} = \boxed{\tfrac{7}{15} \approx 0.467}$$

⋆ *Intuition:* once the conditional law is identified as uniform, every probability question is a ruler measurement on $[0, \text{ceiling}]$ — no new integral, just re-reading Q6's result at a specific x.

Q8 — Uncorrelated Is Not Independent, Reweighted

$X \in \{-1, 0, 1\}$ with $\mathbb{P}(X{=}{-1}) = \mathbb{P}(X{=}1) = 0.25$, $\mathbb{P}(X{=}0) = 0.5$, and $Y = X^2$. Show $\text{Cov}(X,Y) = 0$ yet X, Y are dependent.

Solution. $\mathbb{E}[X] = -1(0.25) + 0(0.5) + 1(0.25) = 0$. $\mathbb{E}[XY] = \mathbb{E}[X^3] = (-1)^3(0.25) + 0 + 1^3(0.25) = 0$.

$$\text{Cov}(X,Y) = \mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y] = 0 - 0 = \boxed{0}$$

Yet Y is *determined* by X: $X{=}0 \Rightarrow Y{=}0$ for certain; $X{=}\pm 1 \Rightarrow Y{=}1$ for certain — the conditional law of Y visibly depends on X, so $\boxed{\text{not independent}}$.

▶ *Intuition:* changing X's probabilities from Example 16's uniform $\frac{1}{3}$ each to 0.25/0.5/0.25 doesn't break the trap — any symmetric distribution of X around 0 kills the *linear* covariance signal with $Y = X^2$ while leaving the deterministic (hence total) dependence fully intact. Memorize the mechanism, not the specific numbers.

Q9 — Correlated Asset Payoffs, Fresh Numbers

$X \in \{-5, 5\}$, $Y \in \{-10, 0, 10\}$ have joint PMF

	$Y{=}-10$	$Y{=}0$	$Y{=}10$
$X{=}-5$	0.20	0.10	0.05
$X{=}5$	0.05	0.15	0.45

Find $\mathbb{E}[Y]$ and $\mathbb{E}[Y \mid X{=}5]$.

Solution. Marginal: $f_Y(-10) = 0.25$, $f_Y(0) = 0.25$, $f_Y(10) = 0.50 \Rightarrow \mathbb{E}[Y] = -10(0.25) + 10(0.50) = 2.5$. $f_X(5) = 0.05 + 0.15 + 0.45 = 0.65$. Conditional: $\left(\frac{1}{13}, \frac{3}{13}, \frac{9}{13}\right)$:

$$\mathbb{E}[Y \mid X = 5] = -10\left(\tfrac{1}{13}\right) + 10\left(\tfrac{9}{13}\right) = \boxed{\tfrac{80}{13} \approx 6.15}$$

versus the unconditional 2.5 — conditioning on Asset A's gain more than doubles the expected Asset B payoff.

⋆ *Intuition:* Example 17's mechanism, re-run with payoffs that can go negative — correlated order flow doesn't care about the sign convention; the same conditioning machinery (restrict to the row, renormalize, take the weighted average) applies unchanged.

Q10 — Full Capstone: Normalize, Marginalize, Test, Condition

$f_{X,Y}(x, y) = c(x + 2y)$ on the unit square $0 < x < 1$, $0 < y < 1$. (a) Find c. (b) Find both marginals and test independence. (c) Find $\mathbb{P}(Y > 0.5 \mid X = 1)$.

Solution. (a) $\int_0^1 \int_0^1 c(x + 2y)\, dy\, dx = c \int_0^1 (x + 1)\, dx = c\left(\frac{3}{2}\right) = 1 \Rightarrow c = \boxed{\frac{2}{3}}$. (b) $f_X(x) = \frac{2}{3}(x + 1)$ on $(0, 1)$; $f_Y(y) = \frac{2}{3}\left(\frac{1}{2} + 2y\right) = \frac{1}{3} + \frac{4}{3}y$ on $(0, 1)$. At $(0, 0)$: $f_X(0) f_Y(0) = \frac{2}{3} \cdot \frac{1}{3} = \frac{2}{9} \neq 0 = f(0, 0) \Rightarrow$ $\boxed{\text{dependent}}$. (c) $f_{Y|X}(y \mid 1) = \dfrac{\frac{2}{3}(1 + 2y)}{\frac{2}{3}(2)} = \dfrac{1 + 2y}{2}$. Then

$$\mathbb{P}(Y > 0.5 \mid X = 1) = \int_{0.5}^{1} \frac{1 + 2y}{2}\, dy = \boxed{\tfrac{5}{8} = 0.625}$$

▶ *Full-circle intuition:* the entire chapter's toolchain on one fresh density — normalize (find c), marginalize (integrate out the other variable), test independence pointwise (one mismatched point convicts), condition (divide joint by marginal), then integrate the conditional to answer a probability question. This is exactly the four-step chain a Stage 1 "hard" joint-distribution question is timed to test.

Drill order for review: re-derive Q3 (why a 0.002 mismatch is still dependence, not rounding noise) and Q8 (state in one sentence why zero covariance never implies independence) closed-book, then Q10 (the full four-stage pipeline) end-to-end. Together they test whether marginals, independence, and conditioning run as one connected skill, not three separate formulas.